For categories, equivalence and isomorphism coincide

The Univalent Foundations Program

hott-book:categories.tex:local-snapshot-20261011; CC BY-SA 3.0 Unported

Editorial note. The counted statement is the author's 'ct:eqv-levelwise' (source lines 812-814, Lemma 9.4.14); the proof environment starts on the next line, so the pairing is by adjacency.

Editorial note. The retained counter values (chapter 9, section 4, theorem offset 13) make the counted lemma print as Lemma 9.4.14, matching the original book numbering taken from the label-number dump of the pinned snapshot.

Editorial note. Two external references point at results of other sections invoked by the proof; they are navigation pointers only, and the counted proof is retained in full.

Editorial note. The retained preamble is the chapter's own retained starter preamble for this pinned source.

Editorial scope. One original result of the HoTT Book, counted once: for categories, equivalence and isomorphism coincide (Lemma 9.4.14).

Allowed standard prerequisites (editorial).

- Equivalence of categories. F is an equivalence of categories when there is G with natural isomorphisms 1_A ~ GF and FG ~ 1_B.. It is the first of the two properties shown equivalent.
- Isomorphism of categories. F is an isomorphism of categories when it is fully faithful and its object map is an equivalence of types.. It is the second property, and the converse direction starts from it.
- Equivalence of precategories versus full faithfulness plus split essential surjectivity. For precategories these two notions agree.. It is the retained result invoked at the end of the converse direction.

Source reference. Original theorem 9.2.5: sources/hott-categories/book-labelnumbers-first-edition.tex, source lines 1103–1103. Original result cited by the retained text but located in another section of the chapter. It is a navigation pointer only: the counted statement's own author proof is retained line for line and nothing is replaced by it. The number is taken from the label-number dump of the pinned book snapshot. It is kept as a pointer rather than a context span because retaining a result from another section as context would print it with this section's number.

Source reference. Original lemma 9.4.5: sources/hott-categories/book-labelnumbers-first-edition.tex, source lines 1100–1100. Original result cited by the retained text but located in another section of the chapter. It is a navigation pointer only: the counted statement's own author proof is retained line for line and nothing is replaced by it. The number is taken from the label-number dump of the pinned book snapshot. It is kept as a pointer rather than a context span because retaining a result from another section as context would print it with this section's number.

Lemma 9.4.14. *For categories A and B, a functor $F : A \to B$ is an equivalence of categories if and only if it is an isomorphism of categories.*

Proof. Since both are mere properties, it suffices to show they are logically equivalent. So first suppose F is an equivalence of categories, with (G, η, ϵ) given. We have already seen that F is fully faithful. By original theorem 9.2.5, the natural isomorphisms η and ϵ yield identities $1_A = GF$ and $FG = 1_B$, hence in particular identities $\mathsf{id}_A = G_0 \circ F_0$ and $F_0 \circ G_0 = \mathsf{id}_B$. Thus, F_0 is an equivalence of types.

Conversely, suppose F is fully faithful and F_0 is an equivalence of types, with inverse G_0, say. Then for each $b : B$ we have $G_0 b : A$ and an identity $FGb = b$, hence an isomorphism $FGb \cong b$. Thus, by original lemma 9.4.5, F is an equivalence of categories. □